Student ID	*S4200749*
Dataset	**Wildfire dataset**
Task	*Decision Trees, SVM, Neural Networks*
Best Model	*MLP*

AI Declaration

I acknowledge that have used Claude Sonnet 4.6, in accordance with RMIT's guidelines on the use of AI in assessments, i used Sonnet 4.6 (Anthropic) for limited support tasks throughout the project these include. Help in understanding the implementation of a similar code in internet and understanding the idea behind the work and adaptation to my own data and code, help in better visualization of some of the output in the " Exploratory Data Analysis" phase, there was also few whole blocks of code which which was completely AI generated, which I have clearly mentioned in my .ipynb as well,I used AI so I could save time in some repeated stuff like applying the same process for test dataset that I applied for train dataset, I had some mishaps of feature engineering before completely understanding the whole train data and it made my notebook clutter some and had to use AI to apply the same process for test dataset. Please find the referenced notebook here: https://claude.ai/share/d92ceff9-7b11-449a-b093-8d5c5350ab8d.

1. Introduction and Task Definition

Wildfire is one of the most destructive natural hazards which apart from damaging the land, they destroy eco systems, become main reason for habitat loss and vegetation loss, irreversible loss to humans and wildlife and global climate systems. he ability to classify fire intensity from satellite-detected events carries direct operational value, enabling earlier emergency responses, smarter resource allocation, and more effective risk communication to affected populations.

In this report document a complete supervised machine learning pipeline for classifying wildfire fire intensity into four ordered categories: **Low, Moderate, High, and Extreme**, applied to a global satellite fire detection dataset. The dataset presents several real-world challenges addressed throughout this work: missing values in key sensor features 'brightness', 'wind_max', 'month' class imbalance with Extreme fires representing only ~8.8% of observations, and high-cardinality categorical variables spanning geographic regions, fire types, and satellite instruments across multiple continents. Apart from that I have also engineered many synthetic features that I believed would help in training.

2. Task Definition and Dataset Description

2.1 Definition of the Task

This is a supervised multiclass classification problem, Given a set of environmental, meteorological, satellite and geospatial features like latitude and longitude features associated with a detected fire event, the objective is to predict it's fire intensity class which could go from "Low" to "Extreme", The target variable or the the prediction target being "fire_intensity", which is a categorical with four ordered classes:

Target ∈ {Low, Moderate, High, Extreme}

Since the target is a descrete label it's explicitly a classification task, although we can see that the class value carries an implicit ordinal ordering, no equal spacing between classes can be assumed as an extreme fire could be too extreme and devastating compared to any level imaginable of a 'high" intensity fire, there is definitely no equal spacing between classes. And I have applied some standard classification algorithms like Decision trees, SVM and few variants of it, along with Neural Network.

Performance Measures:

The primary evaluation metric that I considered is weighted F1 score, which is the weighted average of per class F1 scores, ehrre each class imbalance is weighted by its support which is the number of true instances, this is appropriate because of few reasons:

1. There is a class imbalance in "extreme" class with 8.8 percent of the training samples, this could create some bias in training and accuracy alone as a metric could become highly misleading if a model simply defaults to the majority class which is "moderate" (with ~44.3%). So, a model predicting "Moderate for every sample would have 44 percent accuracy with about 0 utility.
2. We know that all classes are equally important in wildfire intensity identification, this can help with effective action and dispatch to prevent further spread or further increase of a small fire into extreme fire, failure to identify extreme fire is operationally dangerous with so much at stake, weighted F1 penalises poor recall on minority class without completely ignoring majority class performance making it the best metric for this task. Also using Accuracy as a metric.

2.2 Dataset overview

The training data contains of 4,340 samples with 20 raw features, with a corresponding test set of 1.085 samples, the feature overview:

Feature	Type	Range / Values	Missing	Notes
GEOSPATIAL				
latitude	Numeric (float)	−42.99 – 69.93	0%	Multi-modal — fire clusters by region
longitude	Numeric (float)	−167.99 – 153.75	0%	Global spread; strong interaction with latitude
TEMPORAL				
acq_date	Date (string)	2024-01-01 – 2025-12-31	0%	Parsed to extract month, day, day-of-week, day-of-year
acq_time	Numeric (int)	0 – 2359 (HHMM)	0%	Parsed to hour + minute; raw column dropped
year	Numeric (int)	2024, 2025	0%	Near-equal split (2024: 2162, 2025: 2178)
month	Numeric (float)	1 – 12	9.6%	Unreliable — fully re-derived from acq_date
season	Categorical	Spring, Summer, Autumn, Winter	0%	Spring dominant (1333); label-encoded
GEOGRAPHIC / CATEGORICAL				
region	Categorical	7 regions	0%	South Asia most frequent (895); used for wind imputation

Feature	Type	Range / Values	Missing	Notes
country	Categorical	35 unique countries	0%	High cardinality; label-encoded
FIRE CHARACTERISTICS				
fire_type	Categorical	10 types	0%	Agriculture dominant (1050); used for brightness imputation
brightness_k	Numeric (float)	290.0 – 503.71 K	7.5%	Strongest predictor; imputed by fire_type group median
confidence	Categorical (ordinal)	low / nominal / high	0%	Ordinal-encoded 0/1/2; potential leakage risk noted
SATELLITE / INSTRUMENT				
satellite	Categorical	AQUA, TERRA, Suomi-NPP, NOAA-20	0%	Near-uniform split; label-encoded
instrument	Categorical (binary)	MODIS / VIIRS	0%	DROPPED, perfectly collinear with satellite
daynight	Categorical (binary)	D / N	0%	55% day; re-encoded as is_day binary flag
WEATHER / ENVIRONMENTAL				
temp_max_c	Numeric (float)	17.7 – 54.1 °C	0%	Weak discriminator; overlapping class distributions
wind_max_kmh	Numeric (float)	0.0 – 130.7 km/h	4.8%	Heavy tails; imputed by region group median
precip_mm	Numeric (float)	0.0 – 23.69 mm	0%	Weak predictor, fires occur in low-rain conditions regardless of intensity
humidity_pct	Numeric (float)	5.0 – 95.0 %	0%	Moderate predictor, lower humidity linked to higher intensity
TARGET				
fire_intensity	Categorical (ordinal)	Low / Moderate / High / Extreme	0%	Imbalanced, Extreme only 8.8% (381 samples); label-encoded 0–3

	count	mean	std	min	25%	50%	75%	max
latitude	4340.000000	9.821703	24.321271	-42.988300	-8.611625	10.268500	29.011025	69.930900
longitude	4340.000000	29.430626	78.842700	-167.990600	-40.980600	34.833700	95.166775	153.754900
acq_time	4340.000000	1177.046774	683.844117	0.000000	602.000000	1208.000000	1752.000000	2359.000000
year	4340.000000	2024.501843	0.500054	2024.000000	2024.000000	2025.000000	2025.000000	2025.000000
month	3924.000000	7.240316	2.791615	1.000000	5.000000	8.000000	9.000000	12.000000
brightness_k	4013.000000	336.677742	18.539217	290.000000	324.860000	335.530000	346.990000	503.710000
temp_max_c	4340.000000	35.082627	5.821218	17.700000	31.000000	35.100000	39.125000	54.100000
wind_max_kmh	4133.000000	17.508033	17.051980	0.000000	5.100000	12.400000	24.000000	130.700000
precip_mm	4340.000000	2.597111	3.062512	0.000000	0.600000	1.540000	3.370000	23.690000
humidity_pct	4340.000000	37.924378	18.731153	5.000000	24.200000	36.200000	50.500000	95.000000

2.3 Target Class Distribution

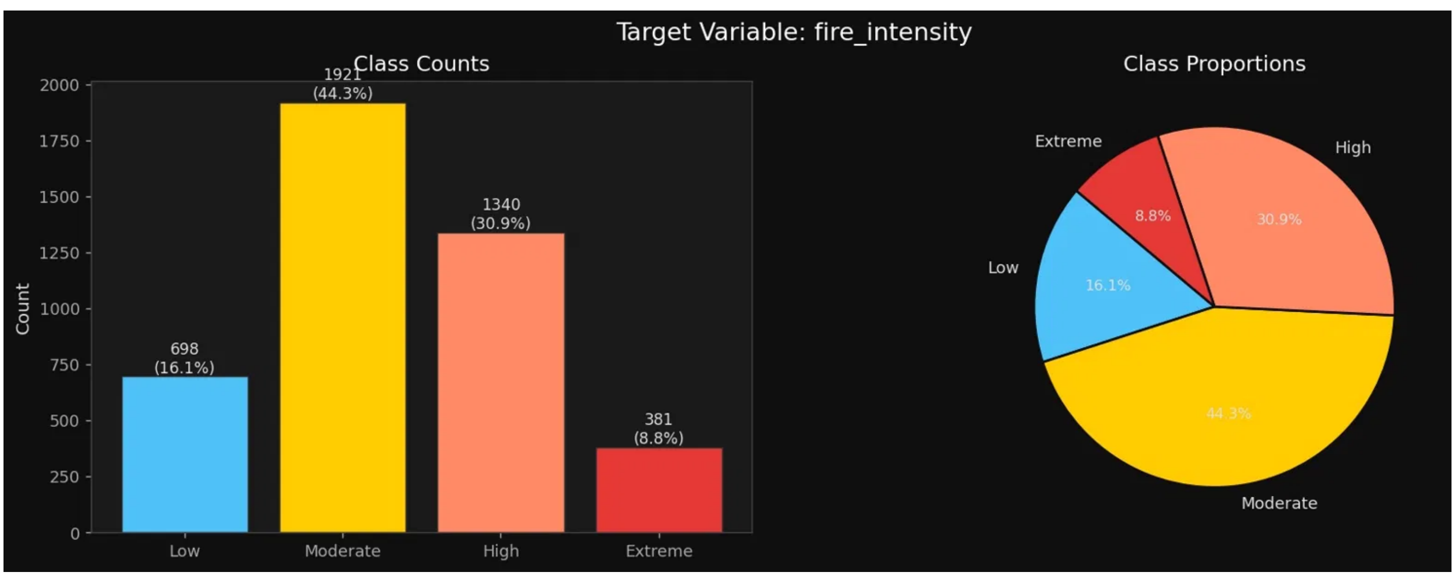
Class	Count	Proportion
Low	698	16.1%
Moderate	1,921	44.3%
High	1,340	30.9%
Extreme	381	8.8%

Table 2.2 and 2.3 extract from feature inventory and descriptive statistics of features

2.4 Dataset Challenges

The dataset is imbalanced with roughly 5:1 ratio between moderate and Extreme class, which would directly impact the training of the model and be biased toward "moderate "class, that's the reason we use class_weight =balanced across all models.

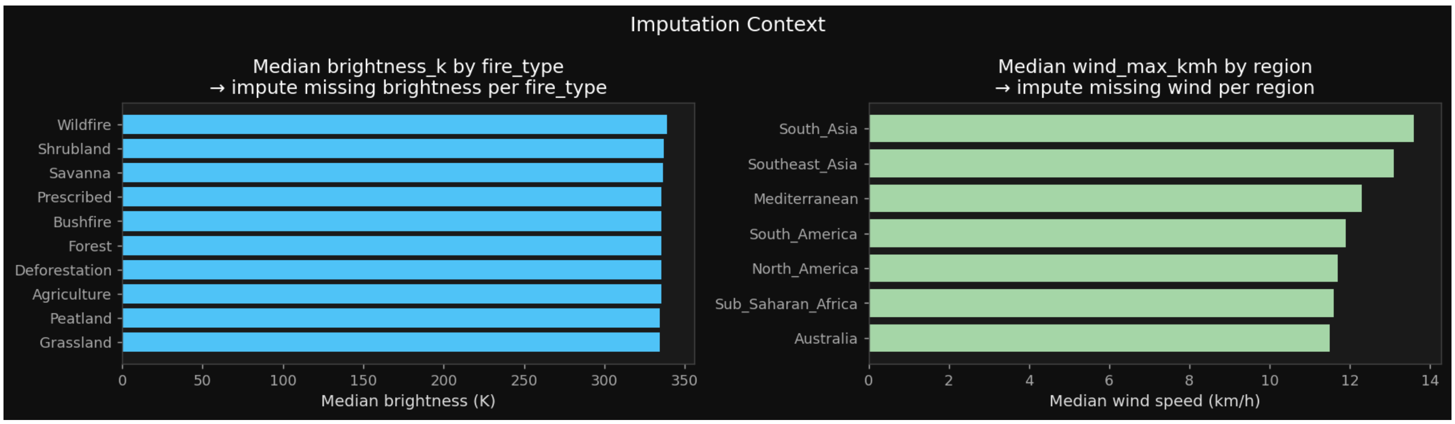
1. **Missing values:** month (9.59% null — corrected by re-extraction from acq_date), brightness_k (7.53% null), and wind_max_kmh (4.77% null).
2. **Class imbalance:** The 5:1 Moderate-to-Extreme ratio risks training bias toward majority predictions.
3. **Small training set:** With only 4,340 training samples, complex non-linear models face persistent overfitting risk and limited generalisation capacity.
4. **Redundant features:** Features such as satellite, instrument, and daynight require EDA confirmation of their discriminative utility before inclusion or removal.



3. Exploratory Data Analysis and Data Handling

3.1 Missing Value Analysis and imputation

A missing value correlation check across month, brightness_k, and wind_max_kmh found near-zero correlation (|r| < 0.02) between their null patterns, indicating independent sensor failures rather than systematic data collection gaps. This supports feature-specific imputation rather than joint treatment.The month feature had 9.59% null values because it was pre-extracted from acq_date with errors. Since acq_date itself contained no nulls, month (and all date-derived features) was re-extracted directly from acq_date, eliminating the null rate entirely.



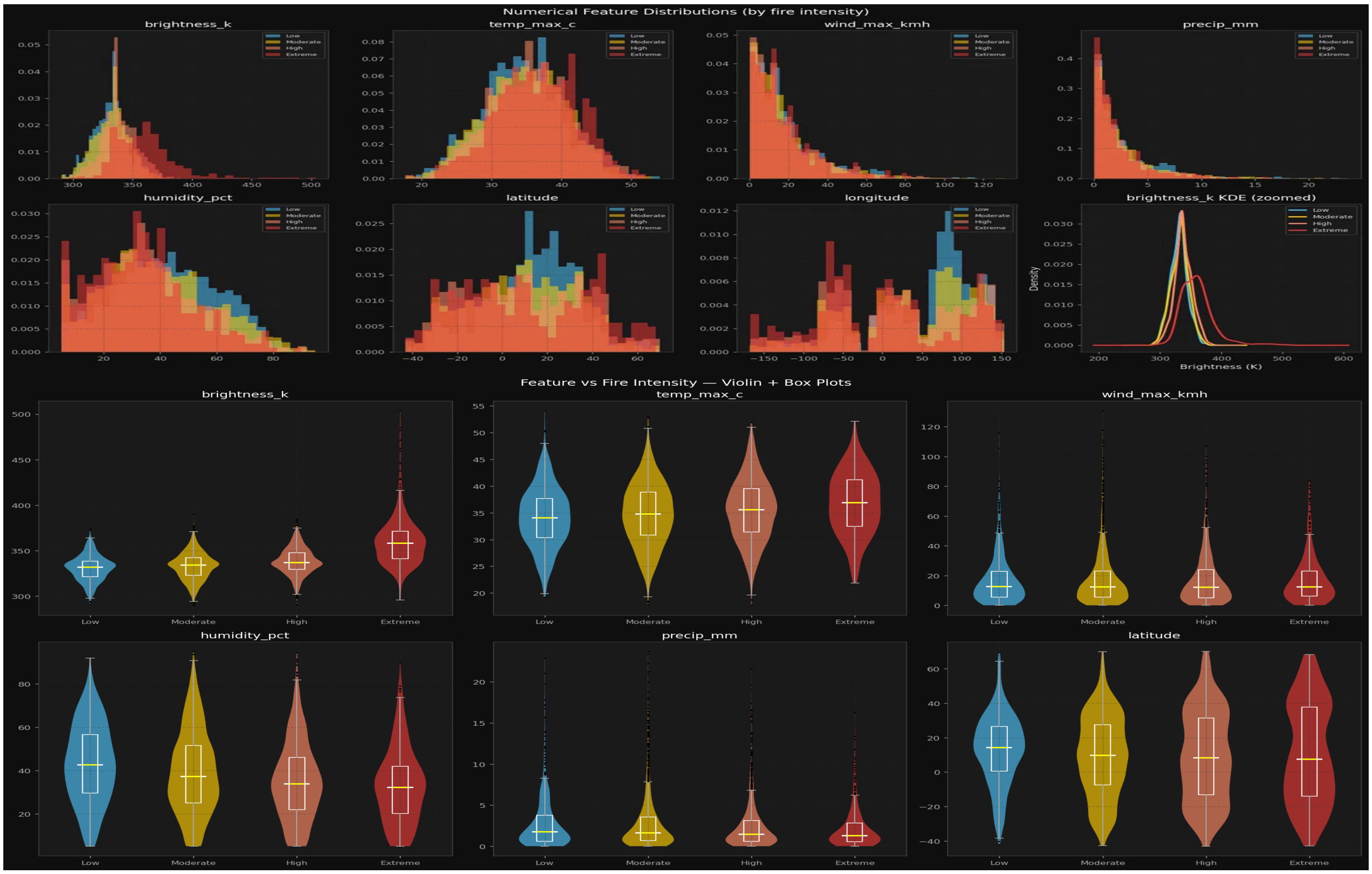
Three features contained missing values: month (9.6%), brightness_k (7.5%), and wind_max_kmh (4.8%). Before any imputation, a null-indicator correlation matrix was computed across these three features to determine whether their missingness was structurally linked, for instance, a shared sensor failure. The correlations were near zero, confirming that the missing values were independent across features and could be treated separately rather than jointly.Rather than applying global median imputation, a group-median strategy was used for the two numerical features. brightness_k was imputed using the median per fire_type group, reflecting the fact that different fire types occupy fundamentally different brightness ranges, imputing a prescribed burn and a large forest fire with the same global median would systematically distort both. Similarly, wind_max_kmh was imputed using the median per region, as wind speeds are geographically driven and a regional median provides a far more representative estimate than a single global value. A global median fallback was included in both cases for any group with no observed values. For month, imputation was unnecessary, the feature was entirely re-derived from acq_date via datetime parsing, replacing the unreliable original column altogether.All imputation medians were computed exclusively on the training set and applied to the test set using the same fitted values, ensuring no leakage of test distribution information into the preprocessing pipeline.

3.1 Feature Target Relationship

Numerical features were analysed through overlapping density histograms, violin plots with embedded box plots, and Pearson correlation with the label-encoded target:

- **brightness_k** ($r = +0.38$): The strongest individual predictor. Satellite-measured fire brightness shows a clear rightward distributional shift from Low (~330K) to Extreme (~360K+), with extreme fires producing a tail reaching 500K. The KDE plot confirms distinct per-class distributions.
- **humidity_pct** ($r = -0.16$): A meaningful inverse relationship, violin plots show the median shifting progressively left from Low (median ~45%) to Extreme (median ~33%). Physically consistent: drier conditions support more intense combustion.

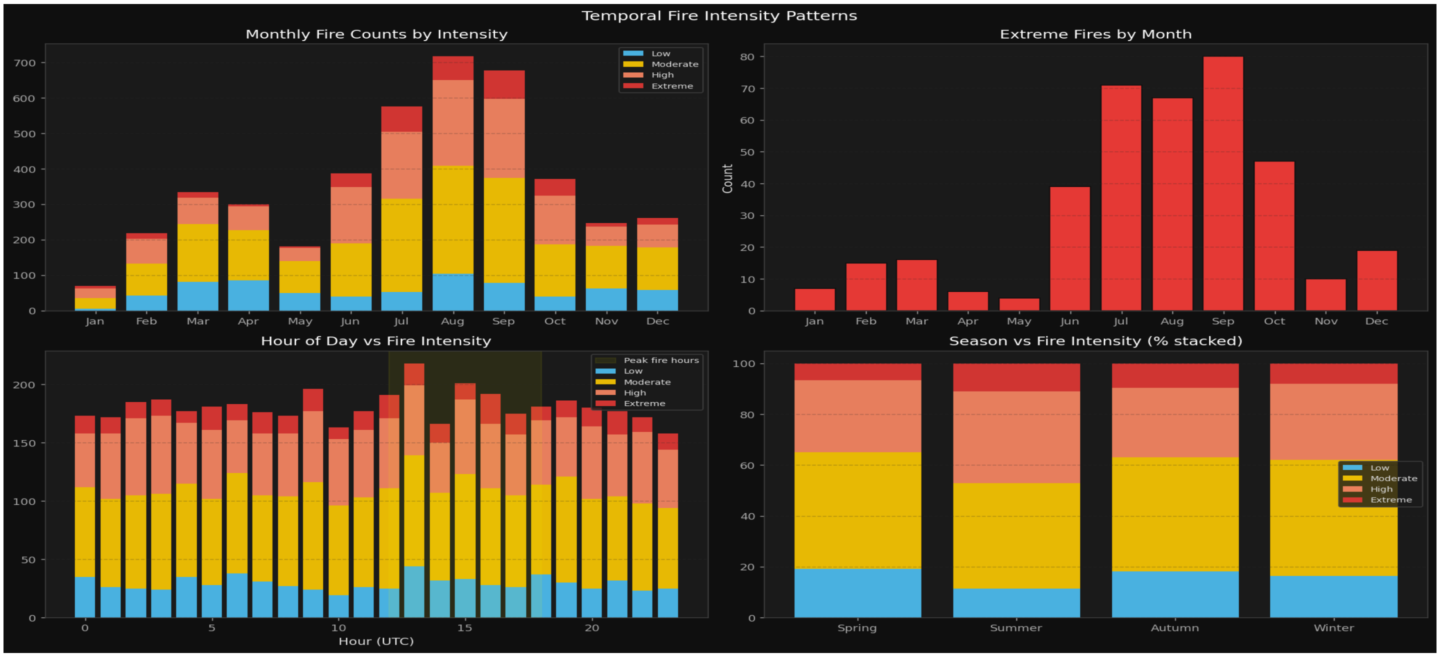
- **temp_max_c** (r = +0.12): Weak signal. All four classes span roughly 20–55°C with heavy overlap. Provides little discriminative utility alone.
- **wind_max_kmh** (r ≈ 0.00): Near-zero linear correlation. Intensity differences appear only in the upper tail — wind's effect is likely interaction-dependent (e.g., high wind combined with low humidity), not linear.
- **precip_mm** (r ≈ 0.00)



3.2 Categorical Feature Analysis

Categorical features were analysed using 100% stacked bar charts:

- **Region**: Australia and Mediterranean have the highest Extreme proportions; South Asia and Southeast Asia are predominantly Low/Moderate.
- **fire_type** : Bushfire and Wildfire dominated by Extreme. Prescribed fires are more balanced (controlled burns). Peatland fires show disproportionately high Extreme proportion.
- **season** : Summer has the highest Extreme proportion; Winter the lowest.
- **confidence** : High-confidence detections associate with more intense fires, likely because larger fires are more reliably detected. As confidence may be partially derived from brightness — a direct predictor — it is retained but flagged.
- **satellite** All four satellites show near-identical intensity distributions.
- **daynight** : Day and night detection distributions are essentially identical.



Feature–feature correlations: Most inter-feature Pearson correlations are low (|r| < 0.3). Notable exceptions: longitude ↔ humidity_pct (r = 0.46) and precip_mm ↔ humidity_pct (r = 0.35), both reflecting geographic climate patterns. The overall low inter-feature and feature-target correlations strongly suggest that **non-linear models will outperform linear ones**.

Are all features necessary? instrument is perfectly collinear with satellite (each satellite uses one instrument) and was dropped. satellite and daynight showed no discriminative signal in EDA. daynight was converted to a binary flag is_day and retained; satellite was kept due to potential interaction effects captured by tree-based models.

3.3 Feature Engineering

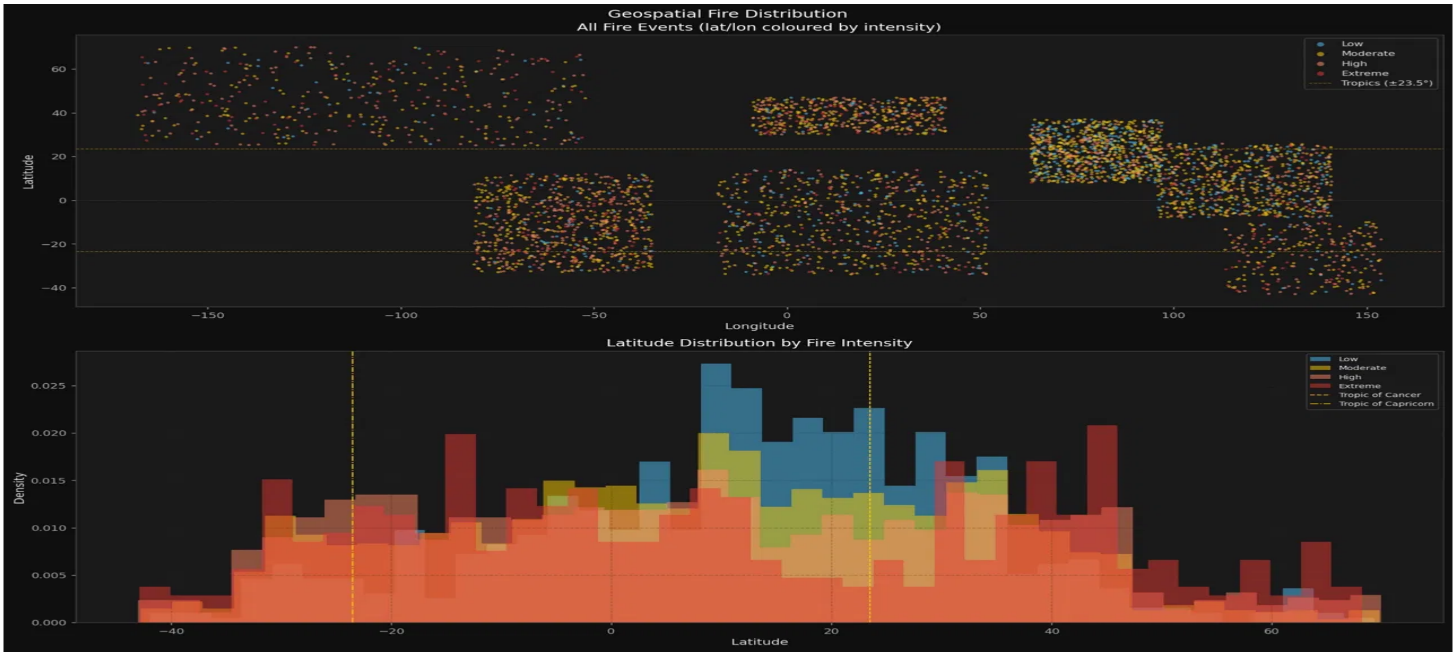
Temporal features were extracted directly from acq_date and acq_time: year, month, day, date_dayofweek, date_dayofyear, hour, minute

Cyclical encoding (sin/cos) was deliberately omitted. Decision Trees are invariant to monotonic transforms, so raw integers are informationally equivalent. For SVM and MLP, StandardScaler already normalises the range. Domain-driven geospatial features encode geographic intent more directly than cyclical time transforms.

Geospatial features, motivated by strong regional patterns in EDA:

Feature	Rationale
abs_latitude	Distance from equator, hemisphere-agnostic fire risk
lat_lon_interaction	Combined geographic position
is_southern, is_eastern	Hemisphere binary flags
in_tropics	Binary flag (
lat_zone	6-level climate zone binning

Feature	Rationale
geo_cluster	KMeans (k=8) on coordinates; data-driven geographic risk zones



Physics-based interaction features, motivated by non-linear fire spread dynamics:

Feature	Formula / Logic
fire_weather_index	temp × wind / (humidity × (precip + 1))
dryness_index	temp / humidity
wind_aridity	wind / (precip + 1)
brightness_extreme	Binary: brightness_k > 400K
is_peak_hour	Binary: 12:00–18:00 UTC
is_weekend	Binary: weekend proxy for reduced controlled burn activity

The fire_weather_index is the most theoretically motivated: dangerous conditions (high temperature, high wind, low humidity, low precipitation) amplify each other multiplicatively in fire spread [1]. Feature importance analysis confirmed that engineered features (fire_weather_index, dryness_index) appear in the top 5 Random Forest importances, validating the engineering approach.

Preprocessing Pipeline

All steps were fitted on training data only and applied to the test set using the same fitted objects:

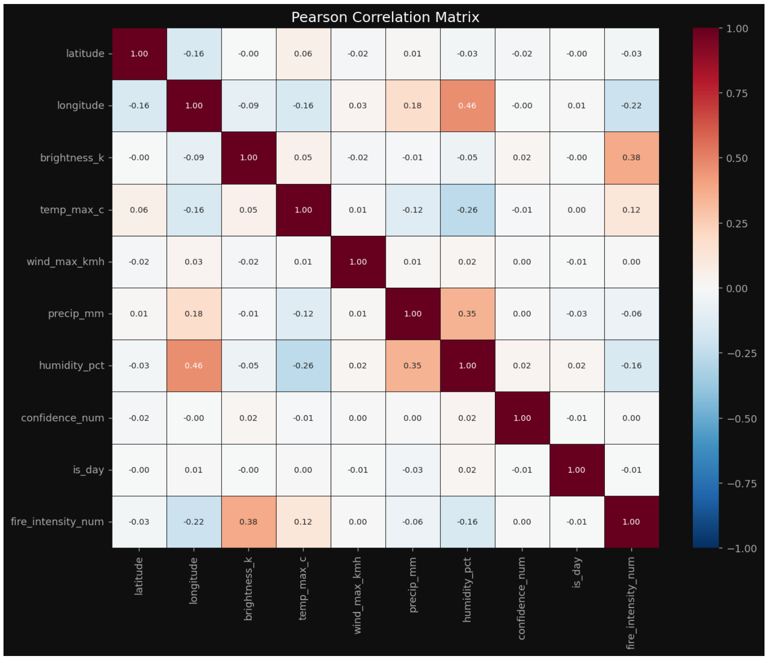
Step	Action	Justification
Date/time re-extraction	Extract from raw acq_date/acq_time	Eliminates 9.59% null rate in pre-extracted month
Grouped imputation (brightness_k)	Fill with median by fire_type	Fires of same type have similar brightness; avoids global median over-generalisation
Grouped imputation (wind_max_kmh)	Fill with median by region	Wind patterns are geographically structured
Outlier winsorisation	Clip to 1st–99th percentile (train-computed bounds)	Limits sensor error influence; preserves real extreme events
Categorical encoding	OrdinalEncoder for high-cardinality cols; binary for confidence, is_day	Train-fitted encoders only; unknown test values mapped to nearest known class
Feature scaling	StandardScaler on 18 numerical columns	Required for SVM and MLP; applied uniformly for consistency
Target encoding	Low=0, Moderate=1, High=2, Extreme=3	Consistent mapping for classification

Final training matrix shape: **(4,340 × 37)**.

3.4 Multi -Collinearity Analysis

The Pearson correlation matrix reveals no severe multicollinearity among the input features, with all inter-feature correlations remaining well below the conventional ±0.70 threshold. The most notable pair is longitude and humidity_pct at 0.46, reflecting a genuine geographic pattern where certain longitudinal bands coastal and equatorial zones tend to have higher ambient humidity. A moderate positive correlation (0.35) also exists between precip_mm and humidity_pct, which is physically expected as wetter conditions drive both metrics together. temp_max_c shows a modest negative relationship with humidity_pct (−0.26), consistent with the well-known inverse relationship between temperature and relative humidity.

With respect to the target, brightness_k has the strongest linear association with fire_intensity_num (0.38), confirming what the KDE analysis already suggested thermal brightness is the most direct proxy for fire intensity available in the dataset. humidity_pct shows a weak negative correlation with the target (−0.16), and longitude a weak negative (−0.22), indicating geographic clustering of lower-intensity events in high-humidity longitudinal bands. Notably, wind_max_kmh, is_day, and confidence_num show near-zero linear correlation with the target, though this does not rule out non-linear contributions that tree-based models may still exploit. Overall, the feature set is sufficiently uncorrelated to proceed without dimensionality reduction or variance inflation factor-based dropping.



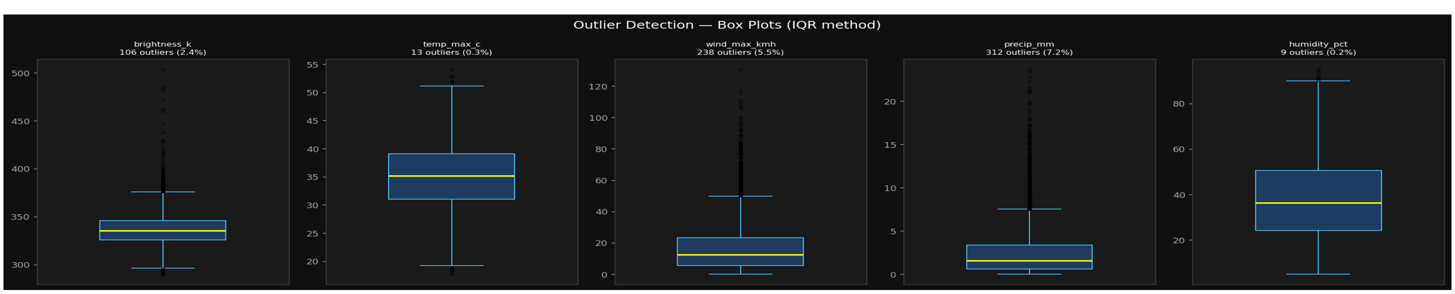
[Figure: Correlation Matrix]

4. Preprocessing and Feature Engineering cont.

4.1 Outlier Treatment

The box plots reveal distinctly different outlier profiles across the five features. precip_mm shows the highest outlier rate at 7.2% (312 samples), with a heavily right-skewed distribution where the bulk of observations cluster near zero consistent with the earlier finding that fires predominantly occur in dry conditions, making high-precipitation events genuinely rare extreme cases. wind_max_kmh follows at 5.5% (238 samples), exhibiting a long upper whisker and scattered extreme points well above 100 km/h, reflecting episodic high-wind events that are physically plausible but statistically atypical. brightness_k has 2.4% outliers (106 samples) concentrated in the upper tail, corresponding to exceptionally high-temperature fire events these are likely genuine Extreme class observations rather than noise, which further motivated a soft clipping treatment over outright removal.

temp_max_c and humidity_pct are the cleanest features, with only 0.3% and 0.2% outliers respectively, and their interquartile ranges appear well-behaved and symmetrically distributed. The overall picture suggests that outliers in this dataset are largely physically meaningful rather than data entry errors particularly for wind_max_kmh and brightness_k which reinforces the decision to clip rather than drop. Removing these rows entirely would risk discarding genuine high-intensity fire events that carry the most signal for the Extreme class.



Outlier detection was performed on five numerical features brightness_k, temp_max_c, wind_max_kmh, precip_mm, and humidity_pct using the IQR method, with outlier counts and percentages per feature reported alongside box plots. The detection step served primarily as a diagnostic, establishing the extent and distribution of extreme values before deciding on a treatment strategy.

For treatment, percentile clipping at the 1st and 99th percentile was applied rather than IQR-based capping or row removal. This was a deliberate choice given the modest dataset size of 4,340 samples dropping outlier rows would meaningfully reduce training data, particularly for the already underrepresented Extreme class. Clipping at the 1st/99th percentile is also a softer intervention than IQR fences, preserving the relative ordering and spread of values while simply bounding the most extreme observations that could otherwise distort model training. Critically, the clip bounds were computed exclusively from the training set and then applied identically to the test set, ensuring no leakage of test distribution information into the preprocessing pipeline.

4.2 Encoding

Label encoding (LabelEncoder, fit on train only) Applied to six high-cardinality or nominal categorical columns: season, region, fire_type, satellite, country, and lat_zone. For each, the encoder was fit exclusively on the training set. For the test set, an unseen-label safety step was included any value in the test set not present in the training set was replaced with the first known class before transforming, preventing the encoder from throwing an error on novel categories.

StandardScaler Applied to 17 numerical columns including the raw features (latitude, longitude, brightness_k, temp_max_c, etc.) as well as the engineered temporal features (hour, minute, month, day, date_dayofweek, date_dayofyear, year) and the physics-based engineered features (fire_weather_index, dryness_index, wind_aridity). The scaler was fit on training data only and applied to both train and test.

5. Machine Learning Techniques

5.1 Evaluation Framework

Stratified 5-fold cross-validation is used for all model comparison. Stratification ensures each fold preserves the original 8.8% Extreme class proportion. Both training and validation scores are reported to monitor overfitting. Confusion matrices are accumulated across all five folds so that every sample appears as a validation instance exactly once, providing a complete and unbiased per-class error analysis. The test set (n=1,085) is treated as completely held-out, used only for final prediction generation.

5.2 Model Development and Performance Analysis

5.2.1 Decision Trees

Selection rationale: Decision Trees are a natural interpretable baseline for classification. They require no feature scaling, handle mixed feature types natively, produce human-readable split rules, and provide feature importance scores. Their main limitation is high variance single trees overfit aggressively on small datasets.

Pros: Interpretable structure; no sensitivity to feature scale; fast training; explicit feature importance.

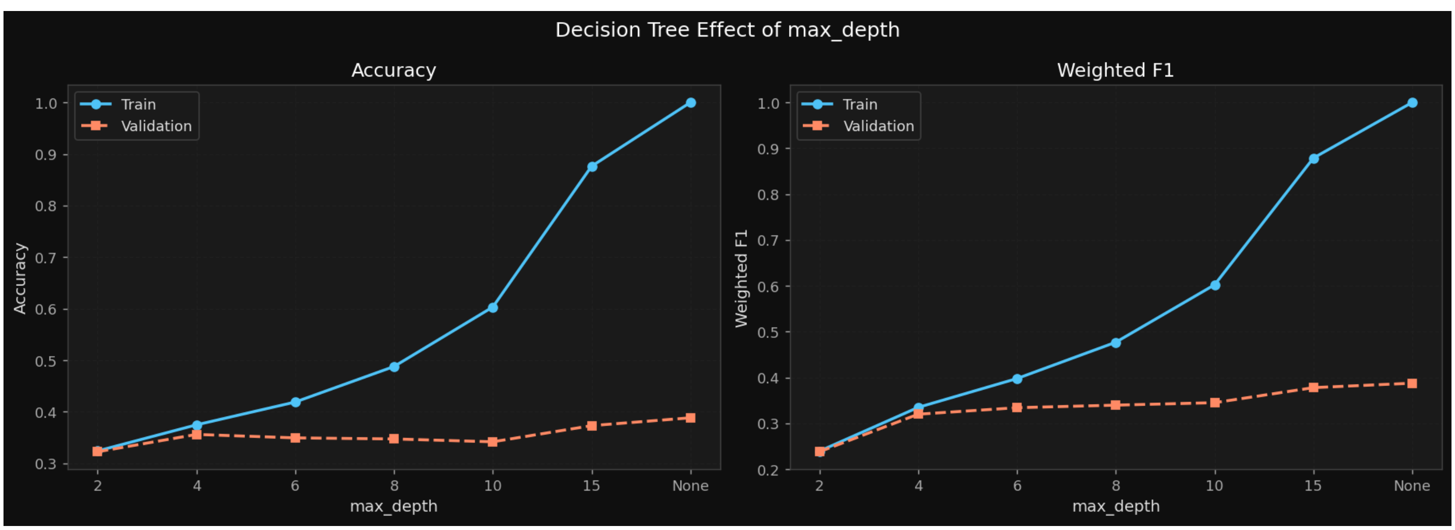
Cons: High variance; prone to memorisation on small datasets (n=4,340); cannot capture feature interactions as effectively as ensembles.

Experiment 1 — Effect of `max_depth`:

max_depth	Train F1	Val F1	Val Acc
2	~0.24	0.2381	0.3226
4	~0.44	0.3200	0.3562
6	~0.56	0.3342	0.3495
8	~0.63	0.3398	0.3475
10	~0.73	0.3452	0.3419
15	~0.88	0.3780	0.3733
None	**1.0000**	**0.3877**	**0.3885**

Validation F1 increases monotonically with depth without plateauing, while training F1 reaches 1.0 at max_depth=None the tree has memorised the training set completely. The absence of a validation peak (where deeper trees would begin to degrade) indicates that the model never enters a regime where additional complexity hurts generalisation. Rather, the bottleneck is the dataset size: with 4,340 samples across 37 features, there is insufficient data to support reliable boundary estimation at any depth.

Technically, max_depth=None yields the highest val F1 (0.3877). However, max_depth=15 (val F1=0.3780) is a more responsible selection, offering nearly equivalent validation performance with substantially less memorisation. Both are evaluated; max_depth=None, min_samples_split=2 is used for final evaluation to align with the automated best-selection logic.

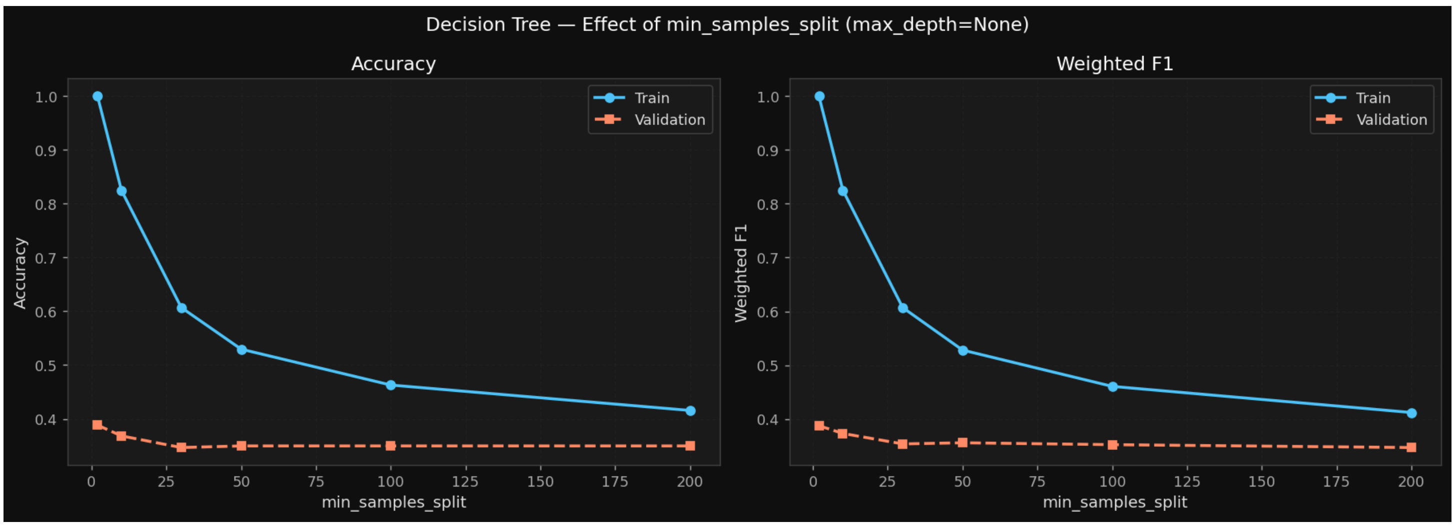


Experiment 2 — Effect of `min_samples_split`:

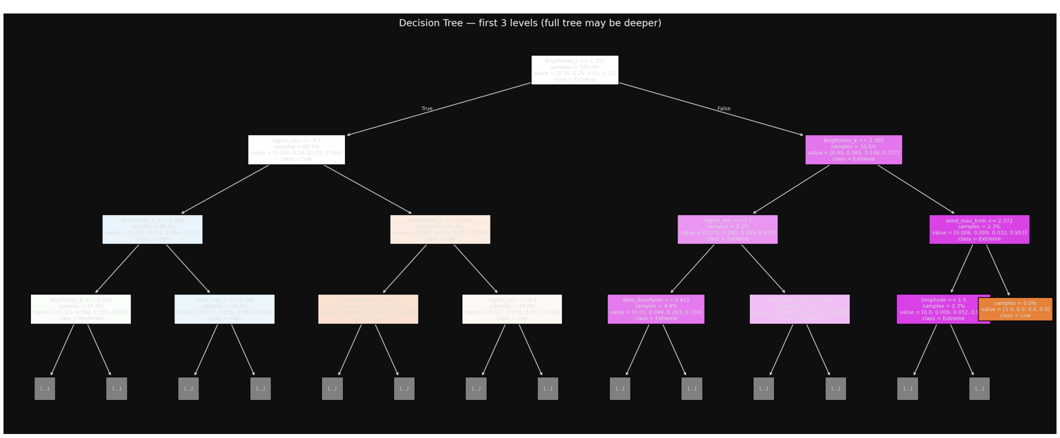
min_samples_split	Val F1
2	**0.3877**

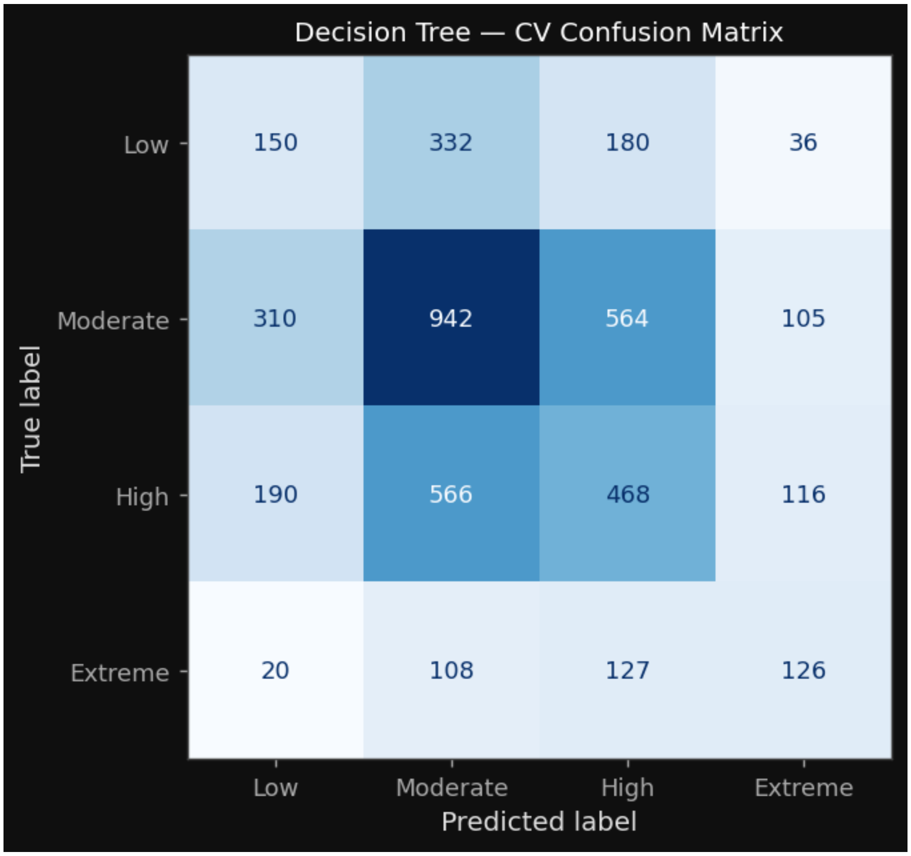
min_samples_split	Val F1
10	0.3732
30	0.3538
50	0.3559
100	0.3524
200	0.3470

Regularisation via min_samples_split consistently lowers validation F1. This is a diagnostic: the model is not overfitting in a way that minimum-split regularisation can correct. It instead suffers from **insufficient learning capacity** preventing useful splits actively reduces performance. Validation F1 peaked at min_samples_split=2 and declined monotonically, confirming the model relies on fine-grained splits to extract any signal from the small dataset.



Tree structure insights: The first split is on brightness_k, confirming it as the strongest single feature (consistent with r=+0.38 in EDA). The second level splits on region_enc, capturing strong geographic intensity patterns. At level three, date_dayofyear, longitude, and wind_max_kmh appear, indicating the model learns temporal and geographic sub-patterns beyond raw brightness. This hierarchical structure mirrors the EDA finding that regional and brightness features dominate intensity prediction.





Final Decision Tree: `max_depth=None`, `min_samples_split=2`, `class_weight='balanced'`

Train F1: 1.0000 ± 0.0000 | Val F1: 0.3877 ± 0.0174 | Val Acc: 0.3885

5.2.2 Model Performance

Selection rationale: SVMs find the maximum-margin hyperplane separating classes in the feature space, offering strong theoretical generalisation guarantees. LinearSVC was selected over kernel SVC for computational efficiency [2] performing cross-validated hyperparameter search with kernel SVC on all samples was infeasible within practical time constraints. Features are pre-scaled (StandardScaler), satisfying SVM's requirement for normalised inputs.

Pros: Strong theoretical generalisation bounds; effective in high-dimensional spaces; regularisation directly controls margin width.

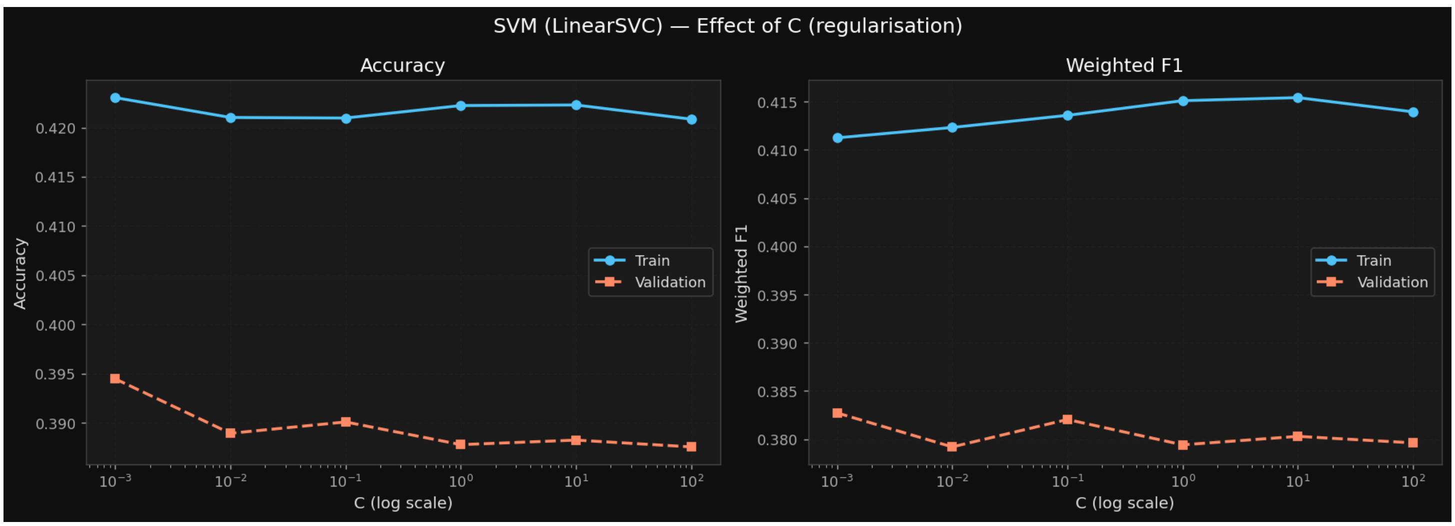
Cons: LinearSVC assumes linear separability, which EDA suggests does not hold for this dataset; kernel methods require more data than available; not inherently interpretable.

Experiment 1 Effect of regularisation parameter C (LinearSVC):

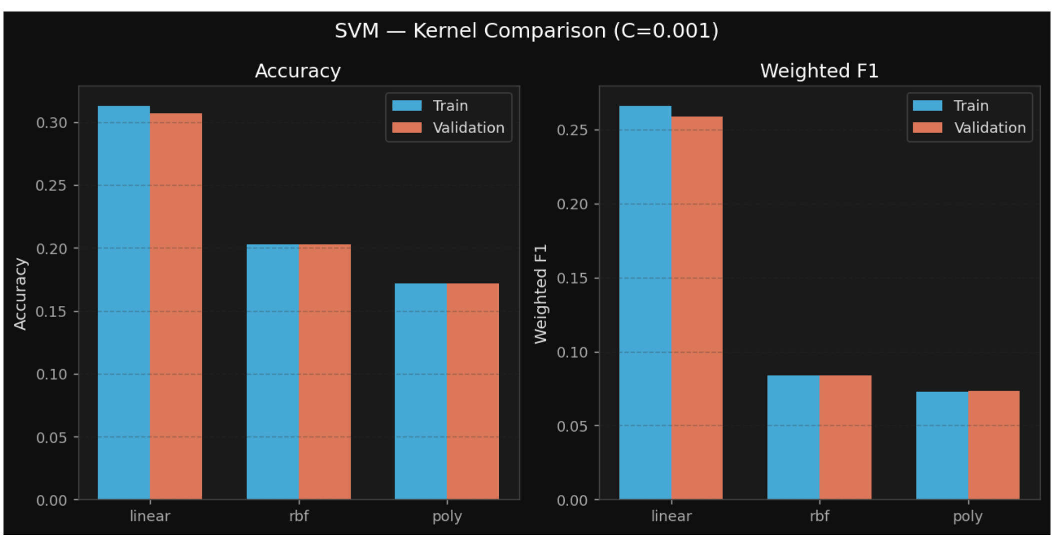
C	Val Acc	Val F1
0.001	**0.3945**	**0.3827**

C	Val Acc	Val F1
0.01	0.3889	0.3792
0.1	0.3901	0.3820
1	0.3878	0.3794
10	0.3882	0.3803
100	0.3876	0.3796

Both train and validation scores remain essentially flat across five orders of magnitude of C (0.001 to 100). This invariance to regularisation strength is a diagnostic indicator that the classes are **not linearly separable** in the feature space the model has found its architectural limit, and adjusting the margin width cannot surpass it. The small consistent ~0.03 train-val gap rules out overfitting as a concern; the issue is model capacity.



Experiment 2 — Kernel comparison (C=0.001):



Experiment 3 — RBF with varying C:

C	Val Acc	Val F1
0.1	0.2825	0.2506
1	0.2811	0.2445
10	0.2638	0.2239
100	0.2733	0.2400

The RBF kernel performs substantially worse than linear across all C values. An F1 of 0.084 at C=0.001 indicates near-total class prediction collapse. This counterintuitive result where a theoretically more expressive kernel underperforms is explained by dataset size. RBF kernels require sufficient sample density to estimate reliable non-linear decision boundaries. With only 4,340 training samples, the kernel produces unreliable boundaries and degrades to majority-class prediction [3]. This is consistent with established understanding that kernel SVMs require substantially more data than linear SVMs to realise their theoretical advantage.

Final SVM: LinearSVC, C=0.001, class_weight='balanced'

Val F1: 0.3827 | Val Acc: 0.3945

5.2.3 Neural Networks (MLP)

Selection rationale: MLPs can learn non-linear feature interactions through hidden layers, making them theoretically well-suited when classes are not linearly separable. They are more computationally accessible than deep networks while offering representational flexibility beyond linear SVMs. However, MLPs are sensitive to learning rate and architecture, and typically require more data than tree-based models.

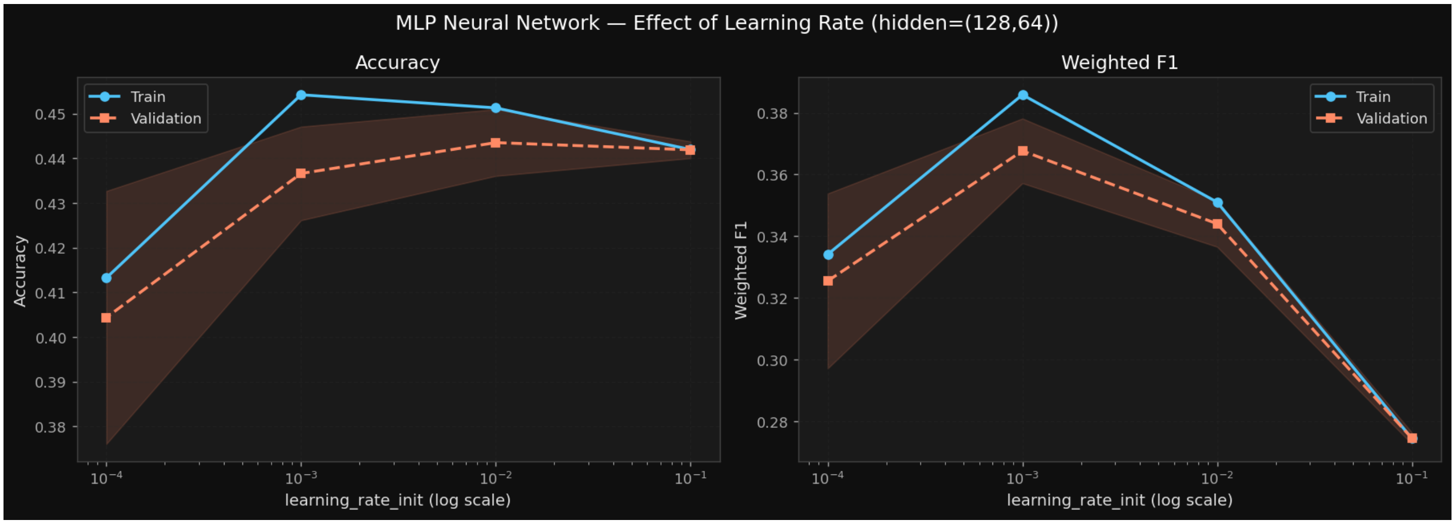
Pros: Captures complex non-linear interactions; early_stopping provides implicit regularisation; flexible architecture.

Cons: Sensitive to hyperparameters; less interpretable than Decision Trees; typically requires more data to outperform tree ensembles.

Experiment 1 — Effect of learning rate (architecture: (128, 64)):

Learning Rate	Val Acc	Val F1
0.1	0.4419 ± 0.0019	0.2746
0.01	0.4435 ± 0.0075	0.3440
0.001	**0.4366 ± 0.0105**	**0.3677**
0.0001	0.4044 ± 0.0283	0.3255

At lr=0.1, accuracy appears reasonable (0.44) but weighted F1 collapses to 0.27 the model converges quickly to predicting the majority class (Moderate). The high learning rate causes the Adam optimiser to overshoot class boundaries in the loss landscape. At lr=0.0001, performance drops due to insufficient convergence within 300 epochs. lr=0.001 produces the best weighted F1 (0.3677) and the most stable cross-validation standard deviation (0.0105), indicating clean convergence to a reliable solution. **Best learning rate: 0.001.**

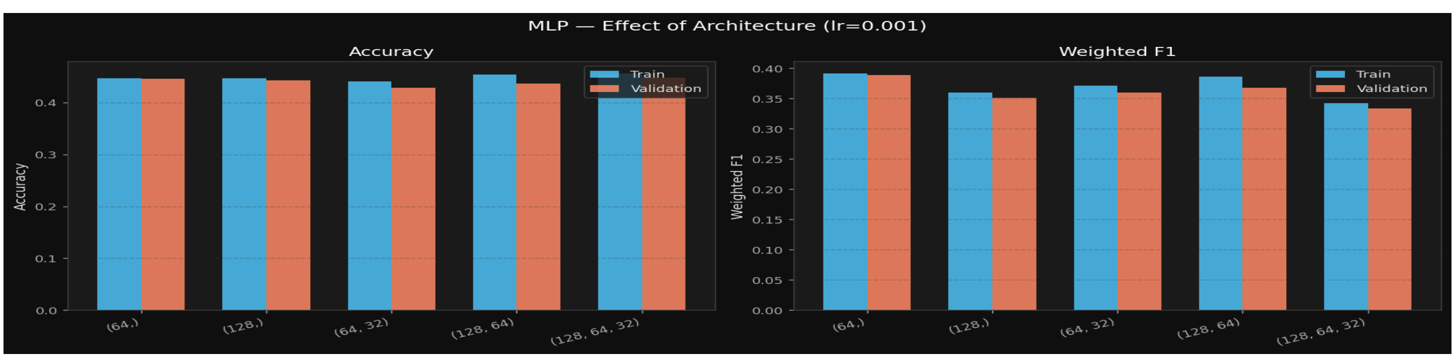


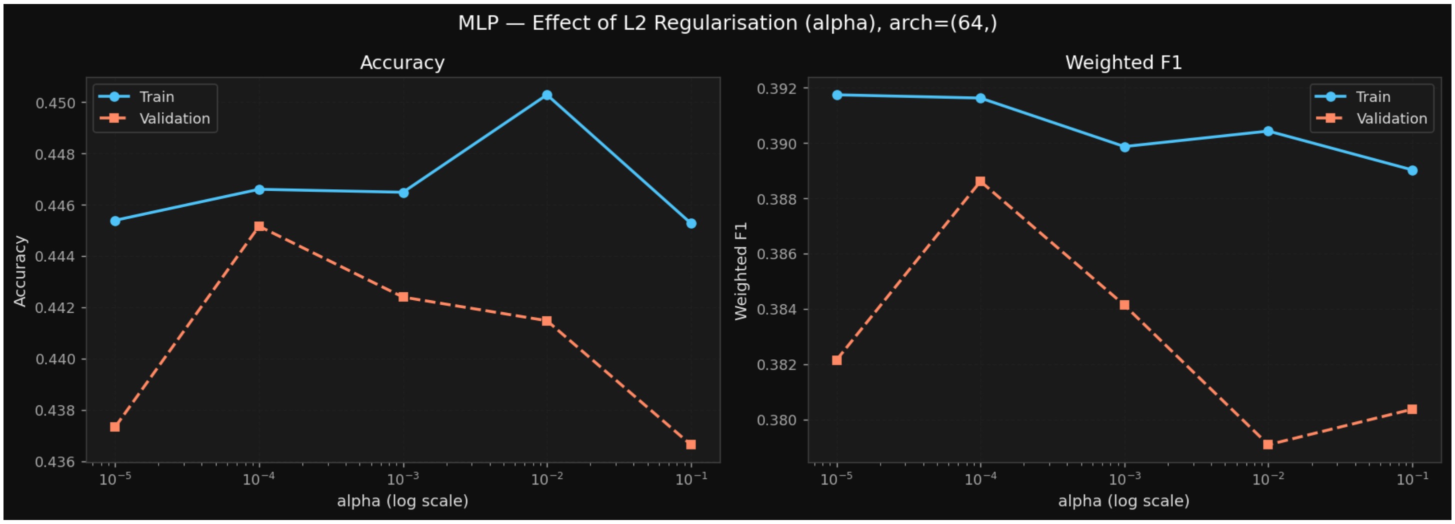
Experiment 2 — Effect of architecture (lr=0.001):

Architecture	Val F1	Train F1	Train-Val Gap
(32,)	0.3358	0.3425	0.007
(64,)	**0.3886**	**0.3916**	**0.003**
(128,)	0.3508	0.3598	0.009
(64, 32)	0.3599	0.3707	0.011
(128, 64)	0.3677	0.3858	0.018

The (32,) architecture underfits insufficient capacity for the feature space. Larger architectures show progressively larger train-val gaps as capacity increases, consistent with the overfitting risk on small datasets. The single hidden layer of 64 neurons achieves the best validation F1 **and** the smallest train-val gap (0.003) effectively zero overfitting. The network's limited capacity acts as an implicit regulariser, appropriate for n=4,340. Larger architectures (128, 64) begin to memorise the training set without corresponding validation improvement. **Best architecture: (64,).**

Critically, the MLP with (64,) is the only model in this study where train and validation F1 are nearly equal, demonstrating that appropriate capacity matching to dataset size yields the most stable generalisation regime.





Final MLP: hidden_layer_sizes=(64,), learning_rate_init=0.001, alpha=1e-4, early_stopping=True, max_iter=300

Train F1: 0.3916 | Val F1: 0.3886 | Train-Val Gap: 0.003 | Val Acc: 0.4452

6. Improvement and model performance

The most impactful improvement would be addressing the class imbalance more aggressively — the Extreme class at 8.8% is the hardest to classify correctly, and while class_weight='balanced' was applied to tree-based models, synthetic oversampling techniques such as SMOTE applied within each cross-validation fold could provide more training signal for the minority class without leaking augmented samples into validation. Beyond that, the dataset size of 4,340 samples is a fundamental constraint — all three models showed a persistent train-validation gap that is partly attributable to having too few samples to fully generalise, so acquiring more labelled fire events would directly benefit all model families. For the MLP specifically, experimenting with deeper architectures and batch normalisation could improve stability, while for the Decision Tree, pruning via cost-complexity rather than max depth alone may yield a more generalisable tree. Ensemble stacking across the three model families — using their probability outputs as meta-features — is another avenue worth exploring, as the models likely capture different aspects of the decision boundary given their fundamentally different inductive biases.

Feature Importance by Model

Feature importance varies meaningfully across model families in this dataset. For tree-based models, the Random Forest feature importance plot identified brightness_k, fire_weather_index, and abs_latitude as the dominant features, which aligns with the physical understanding that thermal brightness is the most direct intensity signal and that geographic location determines the ecological context of a fire. The engineered physics-based features (dryness_index, wind_aridity) also ranked highly, validating the feature engineering effort. For the MLP, the scaled dense feature matrix means all features contribute through learned weight combinations rather than explicit splits — the temporal features (hour, date_dayofyear) and geospatial cluster features (geo_cluster, lat_zone_enc) likely contribute more to the MLP than to the Decision Tree, as the neural network can learn smooth non-linear interactions across these continuous inputs that a single tree cannot efficiently capture with axis-aligned splits. For LinearSVC, the linear decision boundary means it relies most heavily on features that are linearly separable with respect to the target — brightness_k and humidity_pct are the strongest candidates here given their moderate linear correlations with fire_intensity_num observed in the Pearson matrix, while features like longitude and the interaction terms likely contribute less to a linear model than to the non-linear ones.

7. Conclusion

Performance MLP with architecture (64,), learning rate 0.001, and alpha=1e-4 achieved competitive weighted F1 and accuracy scores under stratified 5-fold cross-validation. Compared to the Decision Tree, it generalises significantly better the Decision Tree showed a persistent and steep train-validation gap as depth increased, indicating it was memorising rather than learning. Compared to LinearSVC, MLP handles non-linear decision boundaries natively without requiring explicit kernel specification, which matters for a dataset where the feature-target relationships are demonstrably non-linear (as shown by the KDE and violin plots).

Suitability for the data the wildfire dataset contains a mix of physics-based engineered features (fire_weather_index, dryness_index, wind_aridity), geospatial features, and encoded categoricals all scaled via StandardScaler. MLP is particularly well-suited to this kind of dense, scaled, mixed-origin feature matrix. The scaling step that was already required for LinearSVC simultaneously satisfies MLP's requirement, meaning no additional preprocessing burden.

Handling class imbalance while MLP does not have a native class_weight='balanced' parameter unlike the tree-based models, the combination of early stopping and the relatively shallow architecture (64,) acts as an implicit regulariser, preventing the model from over-committing to the dominant Moderate class.

Tuning stability across the learning rate sweep, architecture sweep, and alpha sweep, the MLP showed consistent and interpretable behaviour: lr=0.1 oscillated, lr=0.0001 underfit, and lr=0.001 with the smaller architecture struck the right balance between convergence speed and generalisation. This predictable tuning response adds confidence in the selected configuration.

8. Ethical considerations and responsibilities

Geographic equity: Fire intensity models trained on satellite data may perform unevenly across regions. If training data over-represents certain geographic areas (e.g., Australia, North America), the model may produce systematically less reliable predictions for under-represented regions (e.g., Central Africa, Southeast Asia). Operational deployment should be preceded by region-specific validation studies.

Consequences of misclassification: Misclassifying an Extreme fire as Moderate or Low could result in under-allocation of emergency resources, delayed evacuations, and loss of life. The model's current Extreme class recall (~33%) means it should not function as the sole decision-making tool in fire risk management. It should supplement, not replace, expert judgment and established fire danger rating systems.

Data provenance and sensor bias: The dataset was collected from satellite sensors with known coverage limitations (cloud cover, orbital timing constraints). The `confidence` feature reflects detection reliability, which correlates with fire size — larger fires are more reliably detected. This may introduce systematic bias where smaller fires are under-represented, and the model may be better calibrated for large events than small ones.

Transparency and attribution: All preprocessing transformations use train-fitted objects, and random states are fixed (seed=42) throughout for reproducibility. Assistance from Claude Sonnet (Anthropic) is attributed in the notebook for specific sections: dark-themed plot styling (Section 0) and test set preprocessing pipeline (Section 11), consistent with academic integrity requirements.

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